# A Survey and Benchmark Evaluation for Neural-Network-Based Lossless Universal Compressors Toward Multi-Source Data (Supplementary Materials)

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December 18, 2024

This document provides detailed information for benchmark datasets and algorithms, as well as additional experiments. The NNLCB is available at `https://github.com/fahaihi/NNLCB`.

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# 1 Benchmark Algorithms Details

## 1.1 NN-Based Compressors

### 1.1.1 Cmix

Cmix [1] is a neural network based lossless compression algorithm aimed at optimizing compression ratio with the cost of high CPU/memory usage. The NNCP compressor uses thousands of context models followed by an NN-based mixer. We used Cmix V19 to finish the experiments.

```
# compression
cmix -c file file.cmix
# decompression
cmix -d file.cmix file.cmix.out
```

### 1.1.2 Lstm-compress

Lstm-compress [2] is an LSTM-based lossless compression algorithm that uses the same LSTM module and preprocessing code as Cmix. Lstm-compress currently only supports compression of a single file. In this manuscript, we used Lstm-compress V3. The detailed commands are as follows.

```
# compression
lstm-compress -c file file.lstm
# decompression
lstm-compress -d file.lstm file.lstm.out
```

### 1.1.3 NNCP

NNCP [3] is a lossless compression algorithm based on LSTM and supports multi-GPU parallel processing. NNCP is an experiment to build a practical lossless data compressor with neural networks. The latest version uses a Transformer model. In this manuscript, we used NNCP V2021-06-01 to finish the experiments. The detailed commands are as follows.

```
# compression
nncp c file file.nncp -T 16 --cuda
# decompression
nncp d file.nncp file.nncp.out -T 16 --cuda
```

### 1.1.4 DeepZip

DeepZip [4] is a general-purpose (universal) compression algorithm based on recurrent neural networks. It belongs to the static pre-training method. The detailed commands for using DeepZip are shown below.

```
# compression
sh ./compress.sh file file.deepzip bs model
# decompression
sh ./decompress.sh file.deepzip file.deepzip.out bs model
```

### 1.1.5 DZip

DZip [5] is an upgraded version of DeepZip, with an extra deeper network added to DeepZip to improve compression. DZip includes two compression modes, combined mode and bootstrap mode. The detailed commands of DZip are as follows.

```
# compression
sh ./compress.sh file file.dzip com model
# decompression
sh ./decompress.sh file.dzip file.dzip.out com model
```

### 1.1.6 TRACE

TRACE [6] is a lossless compression algorithm based on Performer (a Transformer variant.) TRACE uses byte grouping and shared FFNs, and therefore has better execution efficiency. Since the original TRACE puts compression and decompression processes into simultaneous execution, in order to test the performance of compression and decompression separately, we have modified the source files to test the performance of compression or decompression separately.

```
# compression
python compressor.py --source file --comp file.trace
# decompression
python compressor.py --comp file.trace --decomp file.trace.out
```

### 1.1.7 PAC

PAC [7] is a deep learning based compression algorithm fusing MLP and Ordered Mask. Due to the use of MLP, PAC has a lower computational cost. Again, we separated the compression and decompression process of PAC. The running commands of PAC as shown in the command line below.

```
# compression
python compressor.py --source file --comp file.pac
# decompression
python compressor.py --comp file.pac --decomp file.pac.out
```

### 1.1.8 LLMZip

LLMZip [8] uses LLaMA as probabilistic predictor in combination with entropy coding (zlib, Token-by-Token and arithmetic coding) to achieve general-purpose lossless compression. The compression and decompression commands of LLMZip are as follows.

```
# compression
torchrun --nproc_per_node 1 LLMzip_run.py --ckpt_dir llama2/llama-2-7b --tokenizer_path
    llama2/tokenizer.model --win_len 511 --text_file file --compression_folder file
    --encode_decode 0
# decompression
torchrun --nproc_per_node 1 LLMzip_run.py --ckpt_dir llama2/llama-2-7b --tokenizer_path
    llama2/tokenizer.model --win_len 511 --text_file file --compression_folder file
    --encode_decode 1
```

## 1.2 Traditional Compressors

### 1.2.1 GZip

Gzip [9] is a popular early general-purpose lossless compression program originally written by Jean-loup Gailly for the GNU project. The commands for Gzip are shown below.

```
# compression
gzip -c file > file.gz -9
```

```
# decompression
gzip file.gz -9
```

### 1.2.2 PBzip2

PBzip2 [10] is a parallel implementation of the Bzip2 block-sorting file compression algorithm that uses pthreads and achieves near-linear speedup on SMP devices. PBzip2 utilizes the Burrows-Wheeler block sorting algorithm for compressing files, along with Huffman coding for efficient text compression. This manuscript uses parallel Bzip2 V1.1.13 to compress data.

```
# compression
pbzip2 -9 -m2000 -p16 -c file > file.bz2
# decompression
pbzip2 -dc -9 -p16 -m2000 file.bz2
```

### 1.2.3 XZ

XZ Utils [11] is free general-purpose data compression software with a high compression ratio. XZ Utils were written for POSIX-like systems, but also work on some not-so-POSIX systems. XZ Utils are the successor to LZMA Utils. In our experiments, we used XZ V5.5.0. The compression and decompression commands are as follows.

```
# compression
pbzip2 -9 -m2000 -p16 -c file > file.bz2
# decompression
pbzip2 -dc -9 -p16 -m2000 file.bz2
```

### 1.2.4 BSC

BSC [12] is a high-performance file compressor based on lossless block-ordered data compression algorithm, block-ordered data compression algorithm, high-performance file compressor. This manuscript uses BSC V3.3.2 to compress and decompress data.

```
# compression
bsc e file file.bsc -e2
# decompression
bsc d file.bsc file.bsc.out
```

### 1.2.5 SnZip

SnZip [13] is a traditional general-purpose lossless compression algorithm based on snappy. It supports a wide range of file formats including framing-format, old framing-format and so on. The default is framing-format. The command line to run SnZip is as follows.

```
# compression
snzip -k -t snzip file
# decompression
snzip -kd -t snzip file.snz
```

### 1.2.6 Lzma2

Lzma2 [14] improves the multi-threading capability and performance of the LZMA algorithm and better handles incompressible data, so the compression performance is slightly improved. We also used the built-in Lzma2 algorithm in the 7-Zip application.

```
# compression
7zz a -m0=lzma2 -mx9 -mmt16 file.7z file
```

```
# decompression
7zz x -y -mx9 -mmt16 file.7z
```

### 1.2.7 PPMD

PPMD [15] is a context-based compressor, and its core idea is the Partial Matching Prediction (PPM) algorithm proposed by Cleary and Witten. PPM is a statistical modeling technique that uses a set of previous symbols in the input to predict the next symbol to reduce the output data's entropy. PPM differs from a dictionary because PPM predicts the next symbol instead of trying to find the next symbol in the dictionary to encode. We utilized the PPMD in the 7-Zip to compress data.

```
# compression
7zz a -m0=ppmd -mx9 -mmt16 file.7z file
# decompression
7zz x -y -mx9 -mmt16 file.7z
```

### 1.2.8 X3

X3 [16, 17] is a dictionary-based compressor, and encodes various code-stream events using an arithmetic coding. The commands of running X3 are as follows.

```
# compression
x3 -zf file file.x3
# decompression
x3 -df file.x3 file.x3.out
```

### 1.2.9 LZ4-multi

LZ4 [18] is a lossless compression algorithm with compression speeds greater than 500 MB/s per kernel (greater than 0.15 bytes/cycle). Its decoder is extremely fast, up to several GB/s (1 byte/cycle) per kernel. The latest lz4 algorithm supports multi-threaded versions.

```
# compression
lz4 -12 -T16 file file.lz4
# decompression
lz4 -12 -T16 -d file.lz4 file.lz4.reads
```

## 2 Datasets Acquisition

We evaluated the compression performance of all compressors using 19 real datasets with different sizes and types, those open-sourced datasets from literature [6, 19–31]. The detailed information and download hyperlinks of all benchmark datasets are as follows:

Table 1: The Download Linkages of Experimental Datasets

| ID | Dataset | Type | File Size (Byte) | URL |
|---|---|---|---|---|
| D1 | xml [19] | text | 5345280 | Download xml |
| D2 | ooffice [19] | heterogeneous | 6152192 | Download ooffice |
| D3 | reymont [19] | text | 6627202 | Download reymont |
| D4 | sao [19] | homogeneous | 7251944 | Download sao |
| D5 | x-ray [19] | image | 8474240 | Download x-ray |
| D6 | mr [19] | image | 9970564 | Download mr |
| D7 | osdb [19] | heterogeneous | 10085684 | Download osdb |
| D8 | dickens [19] | text | 10192446 | Download dickens |
| D9 | samba [19] | heterogeneous | 21606400 | Download samba |
| D10 | nci [19] | homogeneous | 33553445 | Download nci |
| D11 | webster [19] | heterogeneous | 41458703 | Download webster |
| D12 | mozilla [19] | homogeneous | 51220480 | Download mozilla |
| D13 | enwik8 [20] | text | 100000000 | Download enwik8 |
| D14 | text8 [20] | text | 100000000 | Download text8 |
| D15 | MNIST [24] | image | 54880032 | Download MNIST |
| D16 | CIFAR-10 [25] | image | 186213868 | Download CIFAR-10 |
| D17 | ImageNet [23] | image | 745823247 | Download ImageNet |
| D18 | ImageTest [26] | image | 470611702 | Download ImageTest |
| D19 | Silesia [19] | heterogeneous | 211938580 | Download Silesia |
| D20 | Backup [6] | heterogeneous | 1000000000 | Download Backup |
| D21 | enwik9 [20] | text | 1000000000 | Download enwik9 |
| D22 | Book [21] | text | 1000000000 | Download Book |
| D23 | ESC [22] | audio | 220522000 | Download ESC |
| D24 | Command [27] | audio | 327759206 | Download Command |
| D25 | LibriSpeech [28] | audio | 359034309 | Download LibriSpeech |
| D26 | LJSpeech [29] | audio | 293847664 | Download LJSpeech |
| D27 | DNACorpus [31] | genome | 685597124 | Download DNACorpus |
| D28 | ERR7091247 [30] | genome | 1926041160 | Download ERR7091247 |

# 3 Additional Results

We set the metrics *CS*, *CR*, *CT*, *CPM*, *DT*, *DPM* to denote the compressed file size, compression ratio, compression time, compression peak memory overhead, decompression time and decompression peak memory overhead, respectively.

## 3.1 Detailed Results for Cmix, LLMZip, and X3

### 3.1.1 Cmix

Table 2: Experimental Results Obtained by Running Cmix on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.233 | 0.365 | 114.017 | 18776.344 | 102.917 | 18776.891 |
| D2 | 1.165 | 1.589 | 144.683 | 19457.617 | 142.950 | 19457.723 |
| D3 | 0.674 | 0.853 | 150.567 | 19389.680 | 151.417 | 19389.762 |
| D4 | 3.554 | 4.110 | 167.483 | 19411.430 | 172.650 | 19411.461 |
| D5 | 3.343 | 3.308 | 185.117 | 18827.934 | 195.783 | 18827.828 |
| D6 | 1.747 | 1.469 | 205.983 | 18661.742 | 207.283 | 18661.816 |
| D7 | 1.871 | 1.555 | 235.883 | 19511.012 | 233.017 | 19510.824 |
| D8 | 1.767 | 1.453 | 213.050 | 18689.773 | 203.083 | 18689.563 |
| D9 | 2.392 | 0.928 | 495.650 | 20330.203 | 484.983 | 20335.969 |
| D10 | 0.757 | 0.189 | 545.950 | 18470.406 | 536.733 | 18470.246 |
| D11 | 4.108 | 0.831 | 818.417 | 19809.230 | 826.367 | 19809.277 |
| D12 | 8.394 | 1.374 | 1188.833 | 21832.168 | 1219.567 | 21832.012 |
| D13 | 14.644 | 1.228 | 2026.433 | 22081.254 | 2021.233 | 22081.250 |
| D14 | 15.597 | 1.308 | 1705.917 | 20580.504 | 1701.033 | 20580.387 |
| D15 | 5.361 | 0.819 | 1000.650 | 20360.113 | 999.967 | 20360.168 |

We executed the compression and decompression process of Cmix on 15 datasets of size less than 100 MBs, and the results are shown in Table 2.

From the results, we can see that the execution time and memory overhead of Cmix on the first 15 datasets are much higher than the other algorithms, which is due to the fact that Cmix mixes more than two thousand models to predict the probability distributions of the target symbols, and of course, Cmix outperformed all other algorithms on compression ratio.

### 3.1.2 LLMZip

Table 3: Experimental Results Obtained by Running LLMZip on Benchmark Datasets

| Datasets | *CS* (Bytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| Cmix | | | | | | |
| enwik3 | 262 | 2.096 | 0.084 | 5605.172 | 0.084 | 5605.266 |
| enwik4 | 2797 | 2.237 | 0.268 | 8389.719 | 0.267 | 8389.684 |
| enwik5 | 23799 | 1.903 | 2.067 | 14949.871 | 2.083 | 14949.820 |
| enwik6 | 193830 | 1.550 | 20.451 | 18565.871 | 20.201 | 18566.031 |
| LLMZip | | | | | | |
| enwik3 | 292 | 2.336 | 0.951 | 13700.590 | 0.535 | 13700.773 |
| enwik4 | 668 | 0.534 | 3.618 | 13700.273 | 3.618 | 13701.023 |
| enwik5 | 7653 | 0.612 | 28.952 | 13699.594 | 28.951 | 13700.422 |
| enwik6 | 73855 | 0.591 | 313.550 | 13700.469 | 312.550 | 39246.750 |

LLMZip is a compression algorithm that uses the large language model LLaMa as the probabilistic predictor. Each symbol has to go through the inference operation of the LLM, which also causes it to be much slower than other DL-based and traditional algorithms, and thus cannot complete the benchmark designed in this paper. Therefore, we intercept the first 1,000, 10,000, 100,000, and 1,000,000 bytes of

the English Wikipedia dataset to form the datasets enwik3, enwik4, enwik5, and enwik6 for the test of LLMZip, and select the current algorithm with the optimal compression ratio, Cmix, as the comparison. The experimental results are shown in Table 3.

From the Table 3, we can see that the compression ratio of LLMZip is much better than that of Cmix on the datasets enwik4, enwik5, and enwik6. However, the corresponding computational cost is much higher than that of Cmix, especially the compression/decompression time difference is an order of magnitude.

In conclusion, large language models have great potential for lossless compression, but at the same time the huge computational cost is one of the major challenge.

#### 3.1.3 X3

Table 4: Experimental Results Obtained by Running X3 on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.579 | 0.909 | 3.400 | 22.453 | 2.133 | 22.176 |
| D2 | 2.888 | 3.938 | 14.617 | 105.781 | 10.950 | 106.059 |
| D3 | 1.245 | 1.576 | 2.333 | 27.992 | 1.083 | 27.938 |
| D4 | 5.000 | 5.784 | 5.467 | 100.742 | 2.767 | 100.871 |
| D5 | 4.348 | 4.304 | 8.017 | 59.176 | 5.050 | 59.070 |
| D6 | 2.359 | 1.984 | 4.917 | 53.207 | 2.850 | 53.195 |
| D7 | 3.340 | 2.778 | 6.983 | 51.480 | 3.583 | 51.457 |
| D8 | 2.658 | 2.188 | 6.050 | 42.598 | 3.350 | 42.656 |
| D9 | 5.195 | 2.017 | 89.200 | 163.855 | 57.417 | 163.898 |
| D10 | 1.865 | 0.466 | 5.883 | 53.367 | 3.233 | 53.402 |
| D11 | 8.092 | 1.637 | 143.267 | 177.207 | 98.583 | 176.703 |
| D12 | 18.264 | 2.991 | 1315.317 | 697.516 | 1135.650 | 707.898 |
| D13 | 25.654 | 2.152 | 2260.900 | 575.801 | 1562.167 | 575.262 |
| D15 | 8.946 | 1.367 | 189.333 | 309.422 | 125.183 | 310.305 |
| D16 | 134.085 | 6.040 | 2228.700 | 1237.391 | 1467.433 | 1237.711 |
| D23 | 145.056 | 5.518 | 1430.383 | 1237.141 | 674.700 | 1237.035 |
| D25 | 338.466 | 7.908 | 762.233 | 1134.715 | 364.483 | 1134.820 |
| D26 | 206.139 | 5.885 | 889.767 | 1014.281 | 487.117 | 1014.379 |

X3 is a dictionary-based traditional compression algorithm. Table 4 shows the results of X3 running on only some of the datasets, this is because we tried to execute it multiple times and were unable to complete the test on all datasets on our platform, even with the fastest execution parameters selected. Based on the available results, there is still room for improvement in the compression effectiveness and computational cost of X3.

## 3.2 Detailed Results for Another Algorithms

### 3.2.1 NNCP

Table 5: Experimental Results Obtained by Running NNCP on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.449 | 0.705 | 34.950 | 114.145 | 35.200 | 113.785 |
| D2 | 1.959 | 2.671 | 40.283 | 114.004 | 40.733 | 113.730 |
| D3 | 0.865 | 1.094 | 43.467 | 114.137 | 46.717 | 113.730 |
| D4 | 3.645 | 4.216 | 50.550 | 114.090 | 46.817 | 113.828 |
| D5 | 3.401 | 3.366 | 54.950 | 113.926 | 54.300 | 113.789 |
| D6 | 1.877 | 1.579 | 63.867 | 114.066 | 63.783 | 113.816 |
| D7 | 2.080 | 1.730 | 64.967 | 114.129 | 64.767 | 113.816 |
| D8 | 2.280 | 1.876 | 65.500 | 114.063 | 65.700 | 113.855 |
| D9 | 3.837 | 1.489 | 137.733 | 113.988 | 137.700 | 113.848 |
| D10 | 1.142 | 0.285 | 214.133 | 114.082 | 215.333 | 113.820 |
| D11 | 5.677 | 1.148 | 291.850 | 113.910 | 296.600 | 113.820 |
| D12 | 10.600 | 1.735 | 361.067 | 114.039 | 339.567 | 113.793 |
| D13 | 19.909 | 1.670 | 564.583 | 113.977 | 558.900 | 113.758 |
| D14 | 19.734 | 1.655 | 885.683 | 113.988 | 819.917 | 113.594 |
| D15 | 5.625 | 0.859 | 376.500 | 114.035 | 372.483 | 113.801 |
| D16 | 105.120 | 4.735 | 1231.917 | 114.125 | 1186.133 | 113.629 |
| D17 | 600.826 | 6.757 | 4210.150 | 113.992 | 4197.933 | 113.848 |
| D18 | 183.402 | 3.269 | 3724.267 | 113.992 | 3214.550 | 113.625 |
| D19 | 38.790 | 1.535 | 1325.033 | 114.098 | 1233.300 | 113.793 |
| D20 | 474.456 | 3.980 | 5675.417 | 113.965 | 6593.967 | 113.785 |
| D21 | 156.585 | 1.313 | 6135.417 | 113.949 | 6135.750 | 113.859 |
| D22 | 179.739 | 1.507 | 6001.867 | 114.047 | 5890.100 | 113.719 |
| D23 | 95.072 | 3.616 | 1410.017 | 114.051 | 1459.417 | 113.828 |
| D24 | 136.246 | 3.487 | 2495.900 | 114.051 | 2499.517 | 113.547 |
| D25 | 318.479 | 7.441 | 2895.517 | 113.988 | 2242.367 | 113.715 |
| D26 | 148.944 | 4.251 | 2419.333 | 113.902 | 2258.217 | 113.512 |
| D27 | 144.658 | 1.769 | 4443.283 | 113.965 | 4274.617 | 113.727 |
| D28 | 197.778 | 0.861 | 11357.467 | 114.031 | 11218.567 | 113.770 |

### 3.2.2 Lstm-compress

Table 6: Experimental Results Obtained by Running Lstm-compress on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.455 | 0.713 | 14.867 | 7.199 | 15.533 | 7.199 |
| D2 | 1.964 | 2.677 | 22.967 | 10.145 | 24.817 | 10.082 |
| D3 | 0.956 | 1.210 | 24.550 | 10.082 | 24.100 | 10.172 |
| D4 | 3.769 | 4.359 | 26.350 | 10.168 | 26.717 | 10.141 |
| D5 | 3.407 | 3.372 | 31.467 | 10.070 | 34.083 | 10.227 |
| D6 | 1.876 | 1.578 | 36.350 | 10.191 | 35.750 | 10.191 |
| D7 | 2.111 | 1.755 | 40.467 | 10.137 | 36.967 | 10.070 |
| D8 | 2.334 | 1.920 | 28.183 | 7.180 | 28.650 | 7.184 |
| D9 | 4.090 | 1.587 | 80.033 | 10.141 | 77.400 | 10.168 |
| D10 | 1.460 | 0.364 | 88.517 | 6.590 | 88.400 | 6.633 |
| D11 | 6.100 | 1.234 | 114.050 | 7.230 | 115.617 | 7.277 |
| D12 | 12.416 | 2.033 | 183.783 | 10.168 | 180.867 | 10.191 |
| D13 | 22.097 | 1.854 | 344.950 | 9.238 | 326.800 | 9.168 |
| D14 | 20.936 | 1.756 | 237.233 | 5.867 | 235.067 | 6.023 |
| D15 | 7.170 | 1.096 | 198.800 | 10.176 | 210.667 | 10.125 |
| D16 | 123.055 | 5.543 | 708.683 | 9.969 | 728.567 | 10.164 |
| D17 | 616.534 | 6.934 | 2877.867 | 10.141 | 2790.967 | 10.297 |
| D18 | 405.339 | 7.225 | 2007.267 | 10.066 | 1867.217 | 9.977 |
| D19 | 44.862 | 1.776 | 912.033 | 10.141 | 773.600 | 10.199 |
| D20 | 533.427 | 4.475 | 3728.550 | 10.027 | 3807.933 | 10.074 |
| D21 | 187.254 | 1.571 | 3630.217 | 9.211 | 3577.133 | 9.063 |
| D22 | 203.182 | 1.704 | 3583.200 | 9.113 | 3551.483 | 9.008 |
| D23 | 100.502 | 3.823 | 826.600 | 10.332 | 789.633 | 10.363 |
| D24 | 141.577 | 3.623 | 1224.533 | 9.961 | 1208.200 | 9.992 |
| D25 | 319.552 | 7.466 | 1420.183 | 10.195 | 1316.067 | 10.176 |
| D26 | 156.206 | 4.459 | 1362.583 | 10.008 | 1118.167 | 9.988 |
| D27 | 621.516 | 7.605 | 1398.383 | 5.668 | 1438.817 | 5.598 |
| D28 | 214.544 | 0.934 | 4473.067 | 5.750 | 4505.500 | 5.898 |

### 3.2.3 DeepZip

Table 7: Experimental Results Obtained by Running DeepZip on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.308 | 0.484 | 8.300 | 1952.535 | 1.533 | 1650.543 |
| D2 | 1.560 | 2.127 | 9.717 | 2011.762 | 2.200 | 1675.109 |
| D3 | 0.639 | 0.808 | 10.383 | 1988.621 | 2.217 | 1664.000 |
| D4 | 3.522 | 4.074 | 11.600 | 2020.703 | 2.700 | 1676.785 |
| D5 | 3.377 | 3.343 | 13.500 | 2053.957 | 3.083 | 1698.375 |
| D6 | 1.692 | 1.424 | 15.667 | 2089.121 | 3.417 | 1712.102 |
| D7 | 1.946 | 1.619 | 15.783 | 2062.762 | 3.533 | 1698.344 |
| D8 | 1.910 | 1.572 | 15.867 | 2069.387 | 3.050 | 1693.898 |
| D9 | 3.457 | 1.342 | 34.050 | 2329.758 | 7.333 | 1809.188 |
| D10 | 1.077 | 0.269 | 51.633 | 2608.441 | 8.533 | 1949.051 |
| D11 | 4.955 | 1.003 | 63.550 | 2798.066 | 11.833 | 2009.777 |
| D12 | 10.092 | 1.653 | 80.217 | 3029.996 | 17.867 | 2121.691 |
| D13 | 19.686 | 1.651 | 155.767 | 4112.680 | 33.633 | 2582.793 |
| D14 | 19.349 | 1.623 | 286.967 | 7167.992 | 22.983 | 1701.270 |
| D15 | 5.086 | 0.777 | 149.300 | 4380.559 | 17.783 | 1268.566 |
| D16 | 442.122 | 19.917 | 585.683 | 12398.508 | 98.183 | 2522.895 |
| D17 | 2044.115 | 22.991 | 1413.417 | 18915.352 | 412.217 | 8742.391 |
| D18 | 1133.470 | 20.204 | 815.367 | 30179.367 | 216.483 | 5235.492 |
| D19 | 43.409 | 1.718 | 336.583 | 6691.273 | 73.950 | 3644.801 |
| D20 | 513.489 | 4.307 | 1621.700 | 24736.355 | 390.233 | 11139.488 |
| D21 | 2145.443 | 17.997 | 1738.067 | 24729.777 | 494.533 | 11157.547 |
| D22 | 196.844 | 1.651 | 1605.150 | 24721.637 | 343.817 | 11140.012 |
| D23 | 89.544 | 3.406 | 356.483 | 6867.934 | 79.533 | 3697.969 |
| D24 | 558.719 | 14.300 | 1002.417 | 21055.316 | 149.517 | 3873.039 |
| D25 | 317.534 | 7.419 | 594.250 | 22975.688 | 120.517 | 4171.676 |
| D26 | 725.921 | 20.723 | 522.667 | 19270.680 | 138.333 | 3549.082 |
| D27 | 152.098 | 1.861 | 455.383 | 17525.047 | 95.317 | 8152.754 |
| D28 | 231.633 | 1.009 | 3073.350 | 120284.188 | 392.583 | 19114.172 |

### 3.2.4 DeepZip*

Table 8: Experimental Results Obtained by Running DeepZip* on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 2.836 | 4.450 | 8.300 | 1952.535 | 1.533 | 1650.543 |
| D2 | 5.392 | 7.351 | 9.717 | 2011.762 | 2.200 | 1675.109 |
| D3 | 4.470 | 5.658 | 10.383 | 1988.621 | 2.217 | 1664.000 |
| D4 | 7.353 | 8.506 | 11.600 | 2020.703 | 2.700 | 1676.785 |
| D5 | 7.208 | 7.135 | 13.500 | 2053.957 | 3.083 | 1698.375 |
| D6 | 5.524 | 4.647 | 15.667 | 2089.121 | 3.417 | 1712.102 |
| D7 | 5.777 | 4.805 | 15.783 | 2062.762 | 3.533 | 1698.344 |
| D8 | 4.420 | 3.637 | 15.867 | 2069.387 | 3.050 | 1693.898 |
| D9 | 7.289 | 2.830 | 34.050 | 2329.758 | 7.333 | 1809.188 |
| D10 | 3.417 | 0.854 | 51.633 | 2608.441 | 8.533 | 1949.051 |
| D11 | 7.456 | 1.509 | 63.550 | 2798.066 | 11.833 | 2009.777 |
| D12 | 13.923 | 2.280 | 80.217 | 3029.996 | 17.867 | 2121.691 |
| D13 | 23.264 | 1.952 | 155.767 | 4112.680 | 33.633 | 2582.793 |
| D14 | 21.532 | 1.806 | 286.967 | 7167.992 | 22.983 | 1701.270 |
| D15 | 8.917 | 1.363 | 149.300 | 4380.559 | 17.783 | 1268.566 |
| D16 | 445.953 | 20.089 | 585.683 | 12398.508 | 98.183 | 2522.895 |
| D17 | 2047.946 | 23.034 | 1413.417 | 18915.352 | 412.217 | 8742.391 |
| D18 | 1137.302 | 20.272 | 815.367 | 30179.367 | 216.483 | 5235.492 |
| D19 | 47.240 | 1.870 | 336.583 | 6691.273 | 73.950 | 3644.801 |
| D20 | 517.320 | 4.340 | 1621.700 | 24736.355 | 390.233 | 11139.488 |
| D21 | 2149.027 | 18.027 | 1738.067 | 24729.777 | 494.533 | 11157.547 |
| D22 | 200.407 | 1.681 | 1605.150 | 24721.637 | 343.817 | 11140.012 |
| D23 | 93.375 | 3.552 | 356.483 | 6867.934 | 79.533 | 3697.969 |
| D24 | 562.550 | 14.398 | 1002.417 | 21055.316 | 149.517 | 3873.039 |
| D25 | 321.365 | 7.508 | 594.250 | 22975.688 | 120.517 | 4171.676 |
| D26 | 729.752 | 20.833 | 522.667 | 19270.680 | 138.333 | 3549.082 |
| D27 | 152.225 | 1.863 | 455.383 | 17525.047 | 95.317 | 8152.754 |
| D28 | 233.840 | 1.018 | 3073.350 | 120284.188 | 392.583 | 19114.172 |

### 3.2.5 DZip

Table 9: Experimental Results Obtained by Running DZip on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.284 | 0.446 | 14.767 | 2910.949 | 5.850 | 2503.453 |
| D2 | 1.575 | 2.148 | 17.533 | 2954.473 | 7.783 | 2510.242 |
| D3 | 0.625 | 0.791 | 18.567 | 2965.566 | 7.767 | 2523.012 |
| D4 | 3.519 | 4.071 | 21.117 | 3024.680 | 9.267 | 2515.348 |
| D5 | 3.359 | 3.326 | 24.550 | 3077.281 | 10.800 | 2545.047 |
| D6 | 1.678 | 1.411 | 27.767 | 3193.102 | 11.800 | 2537.941 |
| D7 | 1.931 | 1.606 | 28.267 | 3192.980 | 12.083 | 2547.828 |
| D8 | 1.894 | 1.559 | 28.883 | 3190.648 | 11.317 | 2550.816 |
| D9 | 3.254 | 1.264 | 59.733 | 3912.141 | 26.333 | 2657.793 |
| D10 | 1.054 | 0.264 | 90.300 | 4655.477 | 34.667 | 2737.832 |
| D11 | 4.799 | 0.971 | 112.133 | 5139.852 | 44.750 | 2864.516 |
| D12 | 10.271 | 1.682 | 142.533 | 5725.219 | 61.150 | 2971.930 |
| D13 | 18.774 | 1.575 | 210.300 | 4114.426 | 84.467 | 2830.438 |
| D14 | 18.763 | 1.574 | 344.750 | 7136.996 | 91.600 | 1994.465 |
| D15 | 5.075 | 0.776 | 217.800 | 4368.668 | 90.583 | 1562.145 |
| D16 | 112.007 | 5.046 | 847.700 | 12400.902 | 328.767 | 2816.492 |
| D17 | 672.153 | 7.560 | 1709.917 | 18912.719 | 852.617 | 8974.816 |
| D18 | 197.831 | 3.526 | 1012.050 | 8871.203 | 475.867 | 5527.633 |
| D19 | 39.529 | 1.565 | 472.850 | 6680.633 | 228.783 | 3869.578 |
| D20 | 501.159 | 4.204 | 2236.983 | 24709.680 | 1171.433 | 11394.277 |
| D21 | 179.926 | 1.509 | 2108.517 | 24722.070 | 963.683 | 11396.805 |
| D22 | 186.101 | 1.561 | 2152.317 | 24713.102 | 1073.750 | 11390.449 |
| D23 | 89.188 | 3.393 | 634.200 | 6879.902 | 294.600 | 3954.262 |
| D24 | 144.838 | 3.707 | 1109.917 | 21093.313 | 391.367 | 4167.078 |
| D25 | 317.523 | 7.419 | 803.600 | 22979.719 | 386.367 | 4471.723 |
| D26 | 168.531 | 4.811 | 651.350 | 5667.371 | 310.700 | 3847.301 |
| D27 | 146.872 | 1.797 | 787.133 | 17510.566 | 515.433 | 8402.570 |
| D28 | 206.217 | 0.898 | 4081.667 | 35242.859 | 1398.883 | 19259.707 |

### 3.2.6 DZip*

Table 10: Experimental Results Obtained by Running DZip* on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 2.812 | 4.413 | 14.767 | 2910.949 | 5.850 | 2503.453 |
| D2 | 5.407 | 7.372 | 17.533 | 2954.473 | 7.783 | 2510.242 |
| D3 | 4.456 | 5.640 | 18.567 | 2965.566 | 7.767 | 2523.012 |
| D4 | 7.350 | 8.503 | 21.117 | 3024.680 | 9.267 | 2515.348 |
| D5 | 7.191 | 7.118 | 24.550 | 3077.281 | 10.800 | 2545.047 |
| D6 | 5.509 | 4.635 | 27.767 | 3193.102 | 11.800 | 2537.941 |
| D7 | 5.762 | 4.792 | 28.267 | 3192.980 | 12.083 | 2547.828 |
| D8 | 4.404 | 3.624 | 28.883 | 3190.648 | 11.317 | 2550.816 |
| D9 | 7.086 | 2.751 | 59.733 | 3912.141 | 26.333 | 2657.793 |
| D10 | 3.394 | 0.849 | 90.300 | 4655.477 | 34.667 | 2737.832 |
| D11 | 7.300 | 1.477 | 112.133 | 5139.852 | 44.750 | 2864.516 |
| D12 | 14.102 | 2.310 | 142.533 | 5725.219 | 61.150 | 2971.930 |
| D13 | 22.353 | 1.875 | 210.300 | 4114.426 | 84.467 | 2830.438 |
| D14 | 20.947 | 1.757 | 344.750 | 7136.996 | 91.600 | 1994.465 |
| D15 | 8.906 | 1.361 | 217.800 | 4368.668 | 90.583 | 1562.145 |
| D16 | 115.838 | 5.218 | 847.700 | 12400.902 | 328.767 | 2816.492 |
| D17 | 675.984 | 7.603 | 1709.917 | 18912.719 | 852.617 | 8974.816 |
| D18 | 201.663 | 3.595 | 1012.050 | 8871.203 | 475.867 | 5527.633 |
| D19 | 43.360 | 1.716 | 472.850 | 6680.633 | 228.783 | 3869.578 |
| D20 | 504.990 | 4.236 | 2236.983 | 24709.680 | 1171.433 | 11394.277 |
| D21 | 183.510 | 1.539 | 2108.517 | 24722.070 | 963.683 | 11396.805 |
| D22 | 189.665 | 1.591 | 2152.317 | 24713.102 | 1073.750 | 11390.449 |
| D23 | 93.020 | 3.538 | 634.200 | 6879.902 | 294.600 | 3954.262 |
| D24 | 148.669 | 3.805 | 1109.917 | 21093.313 | 391.367 | 4167.078 |
| D25 | 321.354 | 7.508 | 803.600 | 22979.719 | 386.367 | 4471.723 |
| D26 | 172.363 | 4.921 | 651.350 | 5667.371 | 310.700 | 3847.301 |
| D27 | 146.999 | 1.799 | 787.133 | 17510.566 | 515.433 | 8402.570 |
| D28 | 208.423 | 0.908 | 4081.667 | 35242.859 | 1398.883 | 19259.707 |

### 3.2.7 TRACE

Table 11: Experimental Results Obtained by Running TRACE on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.500 | 0.785 | 3.233 | 2356.668 | 4.733 | 2386.598 |
| D2 | 2.222 | 3.029 | 4.067 | 2358.086 | 5.950 | 2397.297 |
| D3 | 0.938 | 1.187 | 4.100 | 2384.641 | 5.967 | 2398.012 |
| D4 | 3.692 | 4.271 | 4.900 | 2378.813 | 7.233 | 2399.328 |
| D5 | 3.467 | 3.432 | 5.733 | 2384.316 | 8.267 | 2411.211 |
| D6 | 2.000 | 1.682 | 6.383 | 2400.691 | 9.017 | 2411.973 |
| D7 | 2.043 | 1.699 | 6.283 | 2371.844 | 8.900 | 2431.012 |
| D8 | 2.432 | 2.002 | 6.467 | 2384.418 | 9.100 | 2426.105 |
| D9 | 4.539 | 1.762 | 13.417 | 2385.129 | 18.883 | 2513.098 |
| D10 | 1.444 | 0.361 | 18.683 | 2373.801 | 27.717 | 2596.508 |
| D11 | 6.573 | 1.330 | 24.667 | 2373.516 | 35.767 | 2671.699 |
| D12 | 12.292 | 2.013 | 32.367 | 2402.551 | 47.017 | 2711.332 |
| D13 | 22.292 | 1.870 | 41.217 | 1981.125 | 89.783 | 2522.043 |
| D14 | 21.239 | 1.782 | 40.183 | 1967.715 | 73.167 | 2554.129 |
| D15 | 6.251 | 0.955 | 33.733 | 1350.297 | 52.183 | 1824.648 |
| D16 | 114.929 | 5.177 | 124.533 | 3365.840 | 191.350 | 3950.434 |
| D17 | 640.493 | 7.204 | 344.767 | 12451.441 | 713.317 | 12995.129 |
| D18 | 196.117 | 3.496 | 201.000 | 7970.629 | 363.017 | 8564.500 |
| D19 | 44.749 | 1.771 | 83.317 | 3796.121 | 172.833 | 4336.215 |
| D20 | 542.016 | 4.547 | 455.667 | 16573.621 | 1108.517 | 16829.383 |
| D21 | 185.476 | 1.556 | 634.783 | 16570.969 | 1039.200 | 17112.047 |
| D22 | 212.338 | 1.781 | 422.150 | 16572.598 | 889.383 | 17118.887 |
| D23 | 103.291 | 3.929 | 93.933 | 3935.559 | 197.567 | 4476.988 |
| D24 | 143.144 | 3.664 | 143.683 | 5661.016 | 270.150 | 6248.000 |
| D25 | 319.317 | 7.461 | 170.767 | 6167.133 | 289.333 | 6746.480 |
| D26 | 157.156 | 4.486 | 170.133 | 5104.770 | 238.450 | 5696.980 |
| D27 | 152.830 | 1.870 | 290.083 | 11475.219 | 572.033 | 12016.211 |
| D28 | 226.757 | 0.988 | 767.400 | 31565.016 | 1417.767 | 32155.965 |

### 3.2.8 PAC

Table 12: Experimental Results Obtained by Running PAC on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.522 | 0.819 | 3.983 | 2387.141 | 5.417 | 2397.559 |
| D2 | 2.022 | 2.757 | 5.000 | 2374.750 | 6.800 | 2401.742 |
| D3 | 0.960 | 1.215 | 4.817 | 2407.387 | 6.950 | 2408.152 |
| D4 | 3.686 | 4.264 | 5.983 | 2383.949 | 7.900 | 2426.754 |
| D5 | 3.492 | 3.456 | 7.067 | 2380.254 | 8.917 | 2430.973 |
| D6 | 2.129 | 1.791 | 7.750 | 2381.500 | 9.833 | 2431.480 |
| D7 | 2.146 | 1.785 | 7.033 | 2389.156 | 9.883 | 2435.777 |
| D8 | 2.363 | 1.945 | 6.967 | 2387.855 | 10.083 | 2452.203 |
| D9 | 3.969 | 1.541 | 14.467 | 2395.480 | 21.167 | 2537.496 |
| D10 | 1.387 | 0.347 | 21.517 | 2379.309 | 31.233 | 2593.410 |
| D11 | 5.908 | 1.195 | 28.533 | 2362.906 | 39.767 | 2659.699 |
| D12 | 11.156 | 1.827 | 36.617 | 2395.727 | 50.483 | 2732.488 |
| D13 | 20.210 | 1.695 | 51.567 | 1970.391 | 89.450 | 2495.047 |
| D14 | 19.379 | 1.626 | 50.717 | 1970.352 | 85.633 | 2495.000 |
| D15 | 7.239 | 1.106 | 26.033 | 1275.289 | 44.300 | 1747.582 |
| D16 | 99.209 | 4.469 | 103.117 | 3366.324 | 167.433 | 3887.234 |
| D17 | 618.911 | 6.961 | 441.067 | 12441.035 | 671.167 | 12958.430 |
| D18 | 188.614 | 3.362 | 224.750 | 7970.477 | 373.550 | 8493.777 |
| D19 | 40.525 | 1.604 | 108.917 | 3785.918 | 190.683 | 4301.777 |
| D20 | 500.732 | 4.200 | 546.600 | 16561.676 | 863.167 | 17079.383 |
| D21 | 164.170 | 1.377 | 444.583 | 16561.211 | 782.433 | 17084.098 |
| D22 | 189.540 | 1.590 | 491.050 | 16562.176 | 787.517 | 17075.449 |
| D23 | 110.430 | 4.201 | 112.150 | 3924.051 | 0.050 | 372.973 |
| D24 | 151.165 | 3.869 | 150.333 | 5650.383 | 246.450 | 6183.984 |
| D25 | 318.355 | 7.438 | 195.567 | 6161.402 | 306.167 | 6699.219 |
| D26 | 167.597 | 4.784 | 196.117 | 5104.531 | 247.233 | 5623.844 |
| D27 | 147.308 | 1.802 | 300.583 | 11464.469 | 500.250 | 12009.027 |
| D28 | 194.969 | 0.849 | 870.967 | 31566.375 | 1448.133 | 32079.230 |

### 3.2.9 GZip

Table 13: Experimental Results Obtained by Running GZip on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.632 | 0.991 | 0.017 | 1.953 | 0.017 | 1.688 |
| D2 | 2.947 | 4.019 | 0.017 | 1.875 | 0.017 | 1.863 |
| D3 | 1.736 | 2.198 | 0.033 | 1.914 | 0.017 | 1.840 |
| D4 | 5.080 | 5.877 | 0.017 | 1.941 | 0.017 | 1.656 |
| D5 | 5.758 | 5.700 | 0.017 | 1.945 | 0.017 | 1.699 |
| D6 | 3.504 | 2.948 | 0.033 | 1.859 | 0.017 | 1.844 |
| D7 | 3.544 | 2.948 | 0.017 | 1.773 | 0.017 | 1.840 |
| D8 | 3.673 | 3.023 | 0.033 | 2.031 | 0.017 | 1.770 |
| D9 | 5.158 | 2.002 | 0.033 | 1.871 | 0.017 | 1.762 |
| D10 | 2.849 | 0.712 | 0.050 | 1.855 | 0.017 | 1.691 |
| D11 | 11.503 | 2.327 | 0.067 | 1.754 | 0.017 | 1.898 |
| D12 | 18.114 | 2.967 | 0.167 | 1.828 | 0.017 | 1.664 |
| D13 | 34.757 | 2.916 | 0.133 | 1.863 | 0.017 | 1.660 |
| D14 | 31.517 | 2.644 | 0.167 | 1.758 | 0.017 | 1.656 |
| D15 | 10.811 | 1.652 | 0.333 | 1.895 | 0.017 | 1.867 |
| D16 | 162.547 | 7.322 | 0.167 | 1.852 | 0.033 | 1.773 |
| D17 | 702.067 | 7.896 | 0.550 | 1.945 | 0.117 | 1.844 |
| D18 | 330.164 | 5.885 | 0.500 | 1.809 | 0.100 | 1.867 |
| D19 | 64.517 | 2.554 | 0.400 | 1.957 | 0.033 | 1.875 |
| D20 | 743.469 | 6.237 | 0.983 | 1.965 | 0.183 | 1.730 |
| D21 | 307.648 | 2.581 | 1.050 | 1.805 | 0.150 | 1.871 |
| D22 | 342.778 | 2.875 | 1.683 | 1.863 | 0.150 | 1.902 |
| D23 | 156.003 | 5.934 | 0.217 | 1.746 | 0.050 | 1.773 |
| D24 | 225.450 | 5.770 | 0.483 | 1.883 | 0.067 | 1.773 |
| D25 | 321.477 | 7.511 | 0.233 | 1.824 | 0.050 | 1.781 |
| D26 | 240.003 | 6.851 | 0.283 | 1.895 | 0.067 | 1.898 |
| D27 | 177.455 | 2.171 | 9.900 | 1.816 | 0.100 | 1.902 |
| D28 | 370.926 | 1.616 | 9.483 | 1.871 | 0.233 | 1.777 |

#### 3.2.10 PBzip2

Table 14: Experimental Results Obtained by Running PBzip2 on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.418 | 0.656 | 0.017 | 45.313 | 0.017 | 6.902 |
| D2 | 2.726 | 3.716 | 0.017 | 51.441 | 0.017 | 29.777 |
| D3 | 1.188 | 1.504 | 0.017 | 55.328 | 0.017 | 33.676 |
| D4 | 4.713 | 5.452 | 0.017 | 60.984 | 0.017 | 40.992 |
| D5 | 3.864 | 3.824 | 0.017 | 69.965 | 0.017 | 43.566 |
| D6 | 2.343 | 1.971 | 0.017 | 64.012 | 0.017 | 39.199 |
| D7 | 2.674 | 2.224 | 0.017 | 82.059 | 0.017 | 51.309 |
| D8 | 2.669 | 2.196 | 0.017 | 83.395 | 0.017 | 51.625 |
| D9 | 4.363 | 1.693 | 0.017 | 117.348 | 0.017 | 63.711 |
| D10 | 1.766 | 0.441 | 0.017 | 130.473 | 0.017 | 78.910 |
| D11 | 8.248 | 1.668 | 0.017 | 132.527 | 0.017 | 87.598 |
| D12 | 17.066 | 2.794 | 0.017 | 132.715 | 0.017 | 89.582 |
| D13 | 27.671 | 2.321 | 0.017 | 138.285 | 0.017 | 95.387 |
| D14 | 25.178 | 2.112 | 0.033 | 137.379 | 0.017 | 106.250 |
| D15 | 9.580 | 1.464 | 0.017 | 68.574 | 0.017 | 55.383 |
| D16 | 140.276 | 6.319 | 0.033 | 152.461 | 0.033 | 115.441 |
| D17 | 696.059 | 7.828 | 0.117 | 161.840 | 0.083 | 120.848 |
| D18 | 260.493 | 4.643 | 0.083 | 152.891 | 0.050 | 138.977 |
| D19 | 52.120 | 2.063 | 0.050 | 142.414 | 0.017 | 120.184 |
| D20 | 712.635 | 5.978 | 0.133 | 167.500 | 0.083 | 142.363 |
| D21 | 242.298 | 2.032 | 0.117 | 141.875 | 0.067 | 106.531 |
| D22 | 255.441 | 2.142 | 0.117 | 141.996 | 0.067 | 105.223 |
| D23 | 133.081 | 5.062 | 0.050 | 150.148 | 0.033 | 134.223 |
| D24 | 185.766 | 4.754 | 0.067 | 148.301 | 0.033 | 118.242 |
| D25 | 322.556 | 7.536 | 0.083 | 156.027 | 0.050 | 120.512 |
| D26 | 205.676 | 5.872 | 0.067 | 151.281 | 0.033 | 120.789 |
| D27 | 171.842 | 2.102 | 0.083 | 141.438 | 0.067 | 96.395 |
| D28 | 303.885 | 1.324 | 0.200 | 123.426 | 0.083 | 98.289 |

### 3.2.11 XZ

Table 15: Experimental Results Obtained by Running XZ on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.415 | 0.650 | 0.050 | 114.277 | 0.017 | 12.848 |
| D2 | 2.315 | 3.156 | 0.050 | 128.152 | 0.017 | 16.129 |
| D3 | 1.255 | 1.588 | 0.067 | 129.168 | 0.017 | 16.117 |
| D4 | 4.221 | 4.882 | 0.067 | 140.465 | 0.017 | 20.324 |
| D5 | 4.283 | 4.239 | 0.050 | 152.355 | 0.017 | 22.586 |
| D6 | 2.624 | 2.208 | 0.117 | 157.828 | 0.017 | 23.574 |
| D7 | 2.713 | 2.256 | 0.083 | 166.020 | 0.017 | 24.145 |
| D8 | 2.700 | 2.222 | 0.100 | 154.422 | 0.017 | 24.340 |
| D9 | 3.566 | 1.384 | 0.183 | 276.871 | 0.017 | 46.797 |
| D10 | 1.382 | 0.345 | 0.367 | 342.055 | 0.017 | 67.684 |
| D11 | 7.981 | 1.614 | 0.467 | 466.852 | 0.017 | 89.082 |
| D12 | 12.757 | 2.089 | 0.400 | 568.453 | 0.033 | 112.770 |
| D13 | 23.681 | 1.986 | 1.300 | 793.680 | 0.033 | 185.293 |
| D14 | 22.145 | 1.858 | 1.850 | 748.543 | 0.033 | 183.594 |
| D15 | 8.658 | 1.323 | 1.083 | 599.273 | 0.017 | 115.633 |
| D16 | 133.348 | 6.007 | 2.417 | 986.711 | 0.233 | 377.348 |
| D17 | 695.042 | 7.817 | 2.183 | 4100.766 | 0.283 | 1511.156 |
| D18 | 268.436 | 4.785 | 3.067 | 2707.406 | 0.200 | 874.066 |
| D19 | 46.174 | 1.828 | 2.500 | 1079.629 | 0.083 | 320.328 |
| D20 | 605.515 | 5.079 | 2.533 | 4927.270 | 0.250 | 1683.922 |
| D21 | 204.234 | 1.713 | 3.933 | 4525.984 | 0.083 | 1412.191 |
| D22 | 242.129 | 2.031 | 4.867 | 4563.594 | 0.067 | 1456.449 |
| D23 | 135.540 | 5.155 | 1.883 | 1250.883 | 0.150 | 418.043 |
| D24 | 184.560 | 4.724 | 3.100 | 1845.930 | 0.200 | 558.055 |
| D25 | 322.676 | 7.539 | 2.000 | 2013.664 | 0.150 | 753.809 |
| D26 | 205.467 | 5.866 | 2.533 | 1826.195 | 0.233 | 551.094 |
| D27 | 147.148 | 1.800 | 6.050 | 3221.172 | 0.067 | 994.086 |
| D28 | 266.885 | 1.162 | 9.467 | 8231.441 | 0.100 | 2581.180 |

### 3.2.12 BSC

Table 16: Experimental Results Obtained by Running BSC on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.363 | 0.570 | 0.017 | 28.840 | 0.017 | 30.887 |
| D2 | 2.435 | 3.319 | 0.033 | 37.074 | 0.033 | 40.234 |
| D3 | 0.935 | 1.183 | 0.017 | 39.902 | 0.017 | 43.121 |
| D4 | 4.459 | 5.158 | 0.033 | 43.094 | 0.033 | 46.367 |
| D5 | 3.579 | 3.542 | 0.033 | 48.883 | 0.033 | 52.141 |
| D6 | 2.100 | 1.767 | 0.033 | 45.715 | 0.017 | 46.375 |
| D7 | 2.107 | 1.752 | 0.033 | 56.637 | 0.017 | 59.883 |
| D8 | 2.150 | 1.769 | 0.033 | 56.703 | 0.017 | 59.898 |
| D9 | 3.710 | 1.440 | 0.033 | 98.781 | 0.033 | 98.930 |
| D10 | 1.191 | 0.297 | 0.033 | 121.480 | 0.017 | 121.688 |
| D11 | 6.337 | 1.282 | 0.067 | 133.305 | 0.033 | 136.543 |
| D12 | 15.219 | 2.492 | 0.083 | 152.141 | 0.083 | 131.668 |
| D13 | 21.257 | 1.783 | 0.150 | 164.598 | 0.117 | 157.680 |
| D14 | 19.868 | 1.667 | 0.217 | 165.719 | 0.150 | 158.387 |
| D15 | 7.959 | 1.217 | 0.067 | 127.477 | 0.067 | 111.199 |
| D16 | 127.966 | 5.765 | 0.783 | 164.313 | 0.817 | 157.621 |
| D17 | 672.743 | 7.566 | 2.583 | 163.863 | 2.617 | 159.633 |
| D18 | 228.710 | 4.077 | 1.567 | 167.207 | 1.450 | 158.430 |
| D19 | 45.155 | 1.787 | 0.400 | 165.656 | 0.317 | 157.305 |
| D20 | 642.982 | 5.393 | 2.983 | 165.586 | 2.750 | 161.633 |
| D21 | 184.895 | 1.551 | 1.467 | 164.766 | 1.017 | 157.652 |
| D22 | 208.133 | 1.745 | 1.567 | 166.797 | 1.117 | 158.359 |
| D23 | 125.951 | 4.791 | 0.583 | 160.953 | 0.533 | 152.785 |
| D24 | 167.556 | 4.288 | 1.150 | 165.645 | 1.017 | 158.215 |
| D25 | 320.907 | 7.498 | 1.700 | 159.426 | 1.667 | 156.297 |
| D26 | 188.324 | 5.376 | 1.183 | 165.559 | 1.067 | 153.766 |
| D27 | 150.064 | 1.836 | 1.150 | 164.438 | 0.817 | 158.465 |
| D28 | 247.923 | 1.080 | 3.167 | 167.516 | 2.117 | 140.301 |

### 3.2.13 SnZip

Table 17: Experimental Results Obtained by Running SnZip on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 1.240 | 1.945 | 0.017 | 3.141 | 0.017 | 2.949 |
| D2 | 4.092 | 5.580 | 0.017 | 3.230 | 0.017 | 3.152 |
| D3 | 3.057 | 3.869 | 0.017 | 3.156 | 0.017 | 3.152 |
| D4 | 6.140 | 7.102 | 0.017 | 3.078 | 0.017 | 3.266 |
| D5 | 8.067 | 7.985 | 0.017 | 3.223 | 0.017 | 2.980 |
| D6 | 5.150 | 4.333 | 0.017 | 3.316 | 0.017 | 2.941 |
| D7 | 5.079 | 4.224 | 0.017 | 3.086 | 0.017 | 3.133 |
| D8 | 5.960 | 4.906 | 0.017 | 3.152 | 0.017 | 2.965 |
| D9 | 7.645 | 2.968 | 0.017 | 3.219 | 0.017 | 3.109 |
| D10 | 5.841 | 1.460 | 0.017 | 3.285 | 0.017 | 3.063 |
| D11 | 19.062 | 3.857 | 0.017 | 3.309 | 0.017 | 3.117 |
| D12 | 25.342 | 4.150 | 0.017 | 3.113 | 0.017 | 2.984 |
| D13 | 53.351 | 4.475 | 0.033 | 3.086 | 0.033 | 3.133 |
| D14 | 51.297 | 4.303 | 0.033 | 3.102 | 0.033 | 2.961 |
| D15 | 15.626 | 2.388 | 0.017 | 3.070 | 0.017 | 3.164 |
| D16 | 176.024 | 7.930 | 0.017 | 3.172 | 0.017 | 2.984 |
| D17 | 711.190 | 7.999 | 0.017 | 3.172 | 0.017 | 3.105 |
| D18 | 410.976 | 7.326 | 0.050 | 3.113 | 0.033 | 3.109 |
| D19 | 96.736 | 3.829 | 0.050 | 3.223 | 0.033 | 3.164 |
| D20 | 846.365 | 7.100 | 0.100 | 3.277 | 0.050 | 2.980 |
| D21 | 479.171 | 4.020 | 0.300 | 3.129 | 0.150 | 3.211 |
| D22 | 557.717 | 4.678 | 0.333 | 3.203 | 0.183 | 3.090 |
| D23 | 185.278 | 7.048 | 0.017 | 3.113 | 0.017 | 3.211 |
| D24 | 288.635 | 7.387 | 0.033 | 3.230 | 0.017 | 3.203 |
| D25 | 325.677 | 7.609 | 0.017 | 3.113 | 0.017 | 3.109 |
| D26 | 279.587 | 7.982 | 0.017 | 3.230 | 0.017 | 3.043 |
| D27 | 300.928 | 3.682 | 0.217 | 3.195 | 0.133 | 3.078 |
| D28 | 614.493 | 2.676 | 0.417 | 3.293 | 0.267 | 3.121 |

### 3.2.14 Lzma2

Table 18: Experimental Results Obtained by Running Lzma2 on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.434 | 0.680 | 0.033 | 72.598 | 0.017 | 10.652 |
| D2 | 2.313 | 3.154 | 0.033 | 79.758 | 0.017 | 13.430 |
| D3 | 1.257 | 1.591 | 0.050 | 83.848 | 0.017 | 12.637 |
| D4 | 4.209 | 4.869 | 0.033 | 89.227 | 0.017 | 16.246 |
| D5 | 4.272 | 4.229 | 0.033 | 115.832 | 0.017 | 17.355 |
| D6 | 2.621 | 2.205 | 0.050 | 128.418 | 0.017 | 17.141 |
| D7 | 2.720 | 2.262 | 0.050 | 129.332 | 0.017 | 17.586 |
| D8 | 2.700 | 2.222 | 0.067 | 130.445 | 0.017 | 17.613 |
| D9 | 3.586 | 1.392 | 0.083 | 260.414 | 0.017 | 29.402 |
| D10 | 1.661 | 0.415 | 0.133 | 362.805 | 0.017 | 38.707 |
| D11 | 8.000 | 1.618 | 0.267 | 430.871 | 0.017 | 52.746 |
| D12 | 12.726 | 2.084 | 0.200 | 514.559 | 0.017 | 66.824 |
| D13 | 23.711 | 1.988 | 0.850 | 682.023 | 0.017 | 124.434 |
| D14 | 22.149 | 1.858 | 1.583 | 682.176 | 0.033 | 123.070 |
| D15 | 8.785 | 1.343 | 0.333 | 545.938 | 0.017 | 66.285 |
| D16 | 133.627 | 6.020 | 1.150 | 685.051 | 0.117 | 318.336 |
| D17 | 694.413 | 7.810 | 0.667 | 3154.934 | 0.167 | 1422.199 |
| D18 | 268.524 | 4.786 | 1.950 | 1884.727 | 0.183 | 727.109 |
| D19 | 46.455 | 1.839 | 1.217 | 684.805 | 0.050 | 254.590 |
| D20 | 606.015 | 5.083 | 1.133 | 3890.695 | 0.167 | 1575.250 |
| D21 | 204.840 | 1.718 | 2.867 | 3489.426 | 0.067 | 1167.371 |
| D22 | 241.677 | 2.027 | 3.717 | 3525.965 | 0.067 | 1205.047 |
| D23 | 135.099 | 5.139 | 0.833 | 685.141 | 0.117 | 352.543 |
| D24 | 184.310 | 4.717 | 2.650 | 1605.215 | 0.150 | 505.125 |
| D25 | 322.307 | 7.530 | 0.950 | 1832.605 | 0.083 | 675.258 |
| D26 | 204.908 | 5.850 | 1.850 | 1334.871 | 0.183 | 493.852 |
| D27 | 147.151 | 1.800 | 6.700 | 2550.410 | 0.067 | 809.141 |
| D28 | 275.081 | 1.198 | 5.533 | 6614.945 | 0.117 | 2123.047 |

### 3.2.15 PPMD

Table 19: Experimental Results Obtained by Running PPMD on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.343 | 0.538 | 0.017 | 134.531 | 0.017 | 133.602 |
| D2 | 2.381 | 3.245 | 0.033 | 134.969 | 0.033 | 134.219 |
| D3 | 1.005 | 1.272 | 0.017 | 134.969 | 0.033 | 134.141 |
| D4 | 4.562 | 5.277 | 0.050 | 102.859 | 0.050 | 101.992 |
| D5 | 3.673 | 3.635 | 0.050 | 123.859 | 0.050 | 122.844 |
| D6 | 2.213 | 1.862 | 0.033 | 140.684 | 0.033 | 140.055 |
| D7 | 2.274 | 1.891 | 0.033 | 202.895 | 0.033 | 201.977 |
| D8 | 2.175 | 1.789 | 0.050 | 246.914 | 0.050 | 245.766 |
| D9 | 3.639 | 1.412 | 0.050 | 263.074 | 0.067 | 262.262 |
| D10 | 1.416 | 0.353 | 0.033 | 263.176 | 0.050 | 262.293 |
| D11 | 6.621 | 1.339 | 0.117 | 262.930 | 0.133 | 262.141 |
| D12 | 15.175 | 2.485 | 0.150 | 263.105 | 0.167 | 262.336 |
| D13 | 22.099 | 1.853 | 0.350 | 263.180 | 0.367 | 262.348 |
| D14 | 20.420 | 1.713 | 0.467 | 263.395 | 0.483 | 262.375 |
| D15 | 8.563 | 1.309 | 0.200 | 263.203 | 0.200 | 262.367 |
| D16 | 131.200 | 5.910 | 1.700 | 263.199 | 1.833 | 262.371 |
| D17 | 682.264 | 7.673 | 8.083 | 263.410 | 8.167 | 262.410 |
| D18 | 248.956 | 4.438 | 3.333 | 263.426 | 3.600 | 262.195 |
| D19 | 45.882 | 1.816 | 0.750 | 263.434 | 0.850 | 262.297 |
| D20 | 680.304 | 5.706 | 6.867 | 263.168 | 6.150 | 262.273 |
| D21 | 191.069 | 1.602 | 3.633 | 263.297 | 4.100 | 262.418 |
| D22 | 214.733 | 1.801 | 5.150 | 263.109 | 6.483 | 262.383 |
| D23 | 130.039 | 4.946 | 1.400 | 263.156 | 1.450 | 262.223 |
| D24 | 174.001 | 4.453 | 2.483 | 263.211 | 2.850 | 262.227 |
| D25 | 327.188 | 7.645 | 4.450 | 263.441 | 4.933 | 262.246 |
| D26 | 195.713 | 5.587 | 2.833 | 263.383 | 3.133 | 262.359 |
| D27 | 158.915 | 1.944 | 3.817 | 262.926 | 4.133 | 262.406 |
| D28 | 277.780 | 1.210 | 7.433 | 263.387 | 7.733 | 262.246 |

#### 3.2.16 LZ4-multi

Table 20: Experimental Results Obtained by Running LZ4-multi on Benchmark Datasets

| Datasets | *CS* (MegaBytes) | *CR* (bits/base) | *CT* (Minutes) | *CPM* (MegaBytes) | *DT* (Minutes) | *DPM* (MegaBytes) |
|---|---|---|---|---|---|---|
| D1 | 0.726 | 1.140 | 0.017 | 8.109 | 0.017 | 7.445 |
| D2 | 3.374 | 4.600 | 0.017 | 10.984 | 0.017 | 10.680 |
| D3 | 1.971 | 2.494 | 0.017 | 10.602 | 0.017 | 9.973 |
| D4 | 5.408 | 6.256 | 0.017 | 13.508 | 0.017 | 16.223 |
| D5 | 6.848 | 6.778 | 0.017 | 16.301 | 0.017 | 20.172 |
| D6 | 3.998 | 3.364 | 0.033 | 14.094 | 0.017 | 15.234 |
| D7 | 3.774 | 3.139 | 0.017 | 13.820 | 0.017 | 15.078 |
| D8 | 4.182 | 3.442 | 0.017 | 14.332 | 0.017 | 14.613 |
| D9 | 5.825 | 2.262 | 0.033 | 27.215 | 0.017 | 20.438 |
| D10 | 3.460 | 0.865 | 0.017 | 38.273 | 0.017 | 17.145 |
| D11 | 13.209 | 2.673 | 0.017 | 52.863 | 0.017 | 28.012 |
| D12 | 21.015 | 3.442 | 0.050 | 66.531 | 0.017 | 31.414 |
| D13 | 40.043 | 3.359 | 0.033 | 97.676 | 0.017 | 31.473 |
| D14 | 37.157 | 3.117 | 0.033 | 95.270 | 0.017 | 31.055 |
| D15 | 11.989 | 1.832 | 0.200 | 68.129 | 0.017 | 26.840 |
| D16 | 168.490 | 7.590 | 0.033 | 173.766 | 0.017 | 33.363 |
| D17 | 701.494 | 7.890 | 0.067 | 257.613 | 0.017 | 33.832 |
| D18 | 373.514 | 6.658 | 0.117 | 460.441 | 0.017 | 33.605 |
| D19 | 73.797 | 2.921 | 0.067 | 115.992 | 0.017 | 33.125 |
| D20 | 793.085 | 6.653 | 0.117 | 280.258 | 0.033 | 37.563 |
| D21 | 355.190 | 2.980 | 0.167 | 182.148 | 0.033 | 31.461 |
| D22 | 388.988 | 3.263 | 0.217 | 173.086 | 0.033 | 31.516 |
| D23 | 174.370 | 6.633 | 0.033 | 183.230 | 0.017 | 33.410 |
| D24 | 254.529 | 6.514 | 0.067 | 212.379 | 0.017 | 33.820 |
| D25 | 322.461 | 7.534 | 0.033 | 211.270 | 0.017 | 33.488 |
| D26 | 270.723 | 7.728 | 0.033 | 209.367 | 0.017 | 33.434 |
| D27 | 223.475 | 2.734 | 1.133 | 152.094 | 0.033 | 31.066 |
| D28 | 453.821 | 1.977 | 1.200 | 167.094 | 0.050 | 31.004 |

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